

Database Management System Project

Group Id:- T612 Lab Group No:- 6

Topic Name:- e-Bay

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❖ Functional Dependencies and Proof that Relations are in Boyce-Codd Normal Form(BCNF) :-

(1) User_Profile

Attributes:- { Email , Password , first_name , last_name }

Email → Password

Email → first_name

Email → last_name

➤ Let's take the closure of Email:-

$\text{Email}^+ = \{ \text{Email} , \text{Password} , \text{first_name} , \text{last_name} \}$

Candidate-Key:- Email

Primary-Key:- Email

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Email** which is **Candidate-Key** of this relation
Therefore, “User_Profile” is in BCNF

(2) User

Attributes:- { User_id , Email }

User_id → Email

- Let's take the closure of User_id:-
 $\text{User_id}^+ = \{ \text{User_id} , \text{Email} \}$

Candidate-Key:- User_id

Primary-Key:- User_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is **User_id** which is **Candidate-Key** of this relation
Therefore, “User” is in BCNF

(3) Seller

Attributes:- { User_id , item_sold, Avg_Rating , A/C_Number }

User_id → item_sold

User_id → Avg_Rating

User_id → A/C_Number

- Let's take the closure of User_id:-

$\text{User_id}^+ = \{ \text{User_id}, \text{item_sold}, \text{Avg_Rating}, \text{A/C_Number} \}$

Candidate-Key:- User_id

Primary-Key:- User_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is **User_id** which is **Candidate-Key** of this relation
Therefore, "Seller" is in BCNF

(4) Bank_Details

Attributes:- { A/C_Number , Balance }

$\text{A/C_Number} \rightarrow \text{Balance}$

- Let's take the closure of A/C_Number:-

$\text{A/C_Number}^+ = \{ \text{A/C_Number}, \text{Balance} \}$

Candidate-Key:- A/C_Number

Primary-Key:- A/C_Number

- Since in this Minimal Set of FDs, the left side in all of the FDs is **A/C_Number** which is **Candidate-Key** of this relation. Therefore, "Bank_Details" is in BCNF

(5) Product

Attributes:- { Product_id , Product_Name ,
Watching_Number , Available_Units , Price , Description ,
Avg_Rating , Product_Seller_id }

Product_id $\rightarrow$ Product_Name
Product_id $\rightarrow$ Watching_Number
Product_id $\rightarrow$ Available_Units
Product_id $\rightarrow$ Price
Product_id $\rightarrow$ Description
Product_id $\rightarrow$ Avg_Rating
Product_id $\rightarrow$ Product_Seller_id

- Let's take the closure of Product_id:-

$\text{Product_id}^+ = \{ \text{Product_id}, \text{Product_Name}, \text{Watching_Number}, \text{Available_Units}, \text{Price}, \text{Description}, \text{Avg_Rating}, \text{Product_Seller_id} \}$

Candidate-Key:- Product_id

Primary-Key:- Product_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Product_id** which is **Candidate-Key** of this relation.
Therefore, "Product" is in BCNF

(6) Product_Review

Attributes:- { Product_id , User_id , Rating , Comment }

{Product_id , User_id } $\rightarrow$ Rating

{Product_id , User_id } $\rightarrow$ Comments

- Let's take the closure of { Product_id, User_id }:-

$\{ \text{Product_id}, \text{User_id} \}^+ = \{ \text{Product_id}, \text{User_id}, \text{Rating}, \text{Comment} \}$

Candidate-Key:- { Product_id, User_id }

Primary-Key:- { Product_id , User_id }

- Since in this Minimal Set of FDs, the left side in all of the FDs is { **Product_id, User_id** } which is **Candidate-Key** of this relation. Therefore, “**Product_Review**” is in BCNF

(7) Order

Attributes:- { Order_id , Order_Date , Shipping_Cost , Buyer_User_id , Transaction_id ,Shipping_Address_User_id, Total_order_cost }

Order_id → Order_Date

Order_id → Shipping_Cost

Order_id → Buyer_User_id

Order_id → Transaction_id

Order_id → Shipping_User_id

Order_id → Total_order_cost

Transaction_id → Order_id

Transaction_id → Order_Date

Transaction_id → Shipping_Cost

Transaction_id → Buyer_id

Transaction_id → Shipping_User_id

Transaction_id → Total_order_cost

- Let's take the closure of Order_id and Transaction_id:-

$\text{Order_id}^+ = \{ \text{Order_id} , \text{Order_Date} , \text{Shipping_Cost} , \text{Buyer_User_id} , \text{Transaction_id} , \text{Shipping_Address_User_id} , \text{Total_order_cost} \}$

$\text{Transaction_id}^+ = \{ \text{Order_id} , \text{Order_Date} , \text{Shipping_Cost} , \text{Buyer_User_id} , \text{Transaction_id} , \text{Shipping_Address_User_id} , \text{Total_order_cost} \}$

Candidate-Keys:- Order_id , Transaction_id

Primary-Key:- Order_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is Either Order_id or Transaction_id which are **Candidate-Keys** of this relation. Therefore, “Order” is in BCNF

(8) Shipping_Status

Attributes:- { tracking_id , Est_Delivery_Date ,
Delivered_Date , Delivery_Status , Order_id }

tracking_id → Est_Delivery_Date

tracking_id → Delivered_Date

tracking_id → Delivery_Status

tracking_id → Order_id

Order_id → Est_Delivery_Date

Order_id → Delivered_Date

Order_id → Delivery_Status

Order_id → tracking_id

- Let's take the closure of Order_id and tracking_id:-

$\text{Order_id}^+ = \{ \text{tracking_id} , \text{Est_Delivery_Date} , \text{Delivered_Date} ,$
 $\text{Delivery_Status} , \text{Order_id} \}$

$\text{tracking_id}^+ = \{ \text{tracking_id} , \text{Est_Delivery_Date} ,$
 $\text{Delivered_Date} , \text{Delivery_Status} , \text{Order_id} \}$

Candidate-Keys:- Order_id , tracking_id

Primary-Key:- tracking_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is Either Order_id or tracking_id which are **Candidate-Keys** of this relation. Therefore, “Shipping_Status” is in BCNF

(9) Shipper

Attributes:- { Shipper_id , Inv_House_Name , Shipper_Name }

Shipper_id $\rightarrow$ Inv_House_Name

Shipper_id $\rightarrow$ Shipper_Name

- Let's take the closure of Shipper_id:-

$\text{Shipper_id}^+ = \{ \text{Shipper_id} , \text{Inv_House_Name} , \text{Shipper_Name} \}$

Candidate-Key:- Shipper_id

Primary-Key:- Shipper_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Shipper_id** which is **Candidate-Key** of this relation.
Therefore, "Shipper" is in BCNF

(10) Shipper_Phone_No

Attributes:- { Shipper_id , Phone_No }

Phone_No $\rightarrow$ Shipper_id

- Let's take the closure of Phone_No:-

$\text{Phone_No}^+ = \{ \text{Shipper_id} , \text{Phone_No} \}$

Candidate-Key:- Phone_No

Primary-Key:- {Shipper_id , Phone_No}

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Phone_No** which is **Candidate-Key** of this relation.
Therefore, "Shipper_Phone" is in BCNF

(11) Inv_Phone_No

Attributes:- { Inv_House_Name, Inv_Phone_No }

Inv_Phone_No $\rightarrow$ Inv_House_Name

- Let's take the closure of Inv_Phone_No:-

$\text{Inv_Phone_No}^+ = \{ \text{Inv_House_Name} , \text{Inv_Phone_No} \}$

Candidate-Key:- Inv_Phone_No

Primary-Key:- { Inv_House_Name, Inv_Phone_No }

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Inv_Phone_No** which is **Candidate-Key** of this relation.
Therefore, "Inv_Phone_No" is in BCNF

(12) Shipping_Address

Attributes:- { User_id , Apartment_Name , City , Street , State , Pincode , Is_Default }

User_id $\rightarrow$ Apartment_Name

User_id $\rightarrow$ City

User_id $\rightarrow$ Street

User_id $\rightarrow$ State

User_id $\rightarrow$ Pincode

User_id $\rightarrow$ Is_Default

- Let's take the closure of User_id:-

$\text{User_id}^+ = \{ \text{User_id} , \text{Apartment_Name} , \text{City} , \text{Street} , \text{State} , \text{Pincode} , \text{Is_Default} \}$

Candidate-Key:- User_id

Primary-Key:- User_id

- Since in this Minimal Set of FDs, the left side in all of the FDs is **User_id** which is **Candidate-Key** of this relation.
Therefore, “Shipping_Address” is in BCNF

(13) User_Phone

Attributes:- { User_id , Phone_No }

Phone_No $\rightarrow$ User_id

- Let's take the closure of Phone_No:-

Phone_No⁺ = { User_id , Phone_No }

Candidate-Key:- Phone_No

Primary-Key:- {User_id , Phone_No}

- Since in this Minimal Set of FDs, the left side in all of the FDs is **Phone_No** which is **Candidate-Key** of this relation.
Therefore, “User_Phone” is in BCNF

(14) Contains

Attributes:- { User_id , Product_id }

- There does not exist any Functional Dependency
Primary_Key:- { User_id , Product_id }
- Since there does not exist any FD , Therefore the relation
“Contains” is in BCNF

(15) Cart

Attributes:- { User_id }

- There does not exist any Functional Dependency
- Primary_Key:- { User_id }
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(16) Buyer

Attributes:- { User_id }

- There does not exist any Functional Dependency
- Primary_Key:- { User_id }
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(17) Product_Image

Attributes:- {image_url, Product_id }

- There does not exist any Functional Dependency
- Primary_Key:- {image_url, Product_id }
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(18) Has_Category

Attributes:- {Category_Name, Product_id }

- There does not exist any Functional Dependency
- Primary_Key:- { Category_Name ,Product_id }
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(19) Category

Attributes:- { Category_Name }

- There does not exist any Functional Dependency
- Primary_Key:- { Category_Name }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(20) Category_Has_Subcategory

Attributes:- { Category_Name, Subcategory_Name }

- There does not exist any Functional Dependency
- Primary_Key:- { Category_Name, Subcategory_Name }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(21) Subcategory

Attributes:- { Subcategory_Name }

- There does not exist any Functional Dependency
- Primary_Key:- { Subcategory_Name }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(22) Has_Subcategory

Attributes:- { Subcategory_Name, Product_id }

- There does not exist any Functional Dependency
- Primary_Key:- { Subcategory_Name, Product_id }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(23) Payment

Attributes:- { Transaction_id }

- There does not exist any Functional Dependency
- Primary_Key:- { Transaction_id }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(24) Watches

Attributes:- { User_id , Product_id }

- There does not exist any Functional Dependency
- Primary_Key:- { User_id , Product_id }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(25) Has_Order

Attributes:- { Order_id , Product_id }

- There does not exist any Functional Dependency
- Primary_Key:- { Order_id , Product_id }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(26) Delivers

Attributes:- { Shipper_id , Order_id }

- There does not exist any Functional Dependency
- Primary_Key:- { Shipper_id , Order_id }**
- Since there does not exist any FD , Therefore the relation “Contains” is in BCNF

(27) Inv_House

Attributes:- { Inv_House_Name}

- **There does not exist any Functional Dependency**

Primary_Key:- { Inv_House_Name}

- **Since there does not exist any FD , Therefore the relation “Contains” is in BCNF**